The holdout is still untouched

Three pre-registered tests of uncertainty-aware DRO on 72 US ETFs

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Working paper · November 2026

Status: negative results, not a strategy. v1 is closed (H1 and H3 fail, H2 passes). v2 and v3 were stopped by frozen pre-holdout information gates. The 2025–2026 holdout was never touched. This is a methods study on 72 US-listed ETFs; it makes no alpha, stock-selection, causal, or live-deployment claim. Nothing here is investment advice.

Abstract. Does epistemic uncertainty improve the tail-risk/return trade-off of a distributionally robust mean-CVaR portfolio? This paper reports three pre-registered attempts on a point-in-time universe of 72 US ETFs. In v1, a Ridge baseline has no detectable out-of-sample linear signal (−0.019 pooled R^2; mean rank IC 0.013, $t = 0.56$), a mutual-information audit finds no surviving feature, and a reduced uncertainty-aware DRO does not separate from point-estimate DRO: H1 fails, H2 passes, H3 fails. v2 replaces bootstrap sigma with conformalized LightGBM return quantiles; calibration is good (validation coverage 89.9%, test 85.9%), but the uncertainty signal passes only 65.4% of date-level information tests against an 80% gate. v3 replaces return quantiles with a conditional forecast of next-month absolute return; calibration is excellent (4.29% vs 4.30%), but the signal passes only 43.0% and underperforms trailing volatility on development. The contribution is the apparatus: a frozen protocol, a mandatory baseline, an information gate, a non-degenerate comparator, a preserved holdout, and a clean negative result.

1 The question

A standard pipeline predicts expected returns and feeds the point estimates into an optimizer. The optimizer then concentrates in assets whose returns were overestimated — the error-maximization problem. A distributionally robust mean-CVaR optimizer can shrink those bets, and an ML model that reports **where** it is uncertain could, in principle, tell the optimizer which bets to shrink.

That is the hypothesis. It is not the finding.

The project asks a narrow question: does an epistemic-uncertainty layer improve the tail-risk/return trade-off of a DRO mean-CVaR portfolio, net of costs, out-of-sample, relative to point-estimate DRO and simple baselines? The answer is reported under a frozen protocol, with a clean negative result treated as a successful outcome. The project fails only if it is leaky, non-reproducible, or overclaims.

Three attempts were made. Each had a different object of uncertainty. Each was stopped by a gate that was written before the result was seen. The 2025–2026 holdout remains untouched.

2 Apparatus

Data. A frozen snapshot of 72 liquid US-listed equity ETFs, 2000-01 to 2026-09, built from Yahoo adjusted total-return prices, FRED `TB3MS`, and the Ken French three-factor plus momentum series. The universe is point-in-time: an ETF enters only on or after its provider first-trade date, and eligibility requires at least 36 months of history, a $5 price floor, and $1M average daily dollar volume. Breadth is 68+ ETFs in every 2010+ month. The snapshot is current-listed, so closed funds are omitted; no survivorship-free claim is made.

Features. Nineteen lagged ETF features: momentum at 1/3/6/12 months, volatility at 3/12 months, downside volatility, skew, worst month, volatility ratio, volume shock, 36-month market correlation, 12-month drawdown, dollar volume, 36-month beta, and four factor exposures. Every feature at month-end t uses only data timestamped at or before t. The output is cross-sectionally winsorized and ranked, with complete cases only.

Protocol. Monthly walk-forward. A 60-month training window, a 12-month validation window, a one-month purge, and a 12-month test window. No random splits. Long-only, sum to one, 5% cap, 10 bps one-way costs as

the primary case, with 5/20/50 bps sensitivity. Net Sharpe is the primary metric; max drawdown and CVaR95 are guardrails. The solver is CLARABEL; a solver failure is fatal, with no equal-weight fallback.

Gates. Five rules did most of the work:

1. a mandatory Ridge B0 baseline before any complex model;
2. a date-blocked mutual-information audit with a permutation null;
3. a target-shuffling falsification test;
4. a counter-metric gate requiring Sharpe, max drawdown, and CVaR95 to pass together; and
5. a pre-holdout information gate for any new uncertainty signal.

The last gate is why the holdout survived.

3 v1: epistemic uncertainty

The first attempt used the original design: Ridge B0 mean, a five-model bootstrap uncertainty proxy, and a box DRO with a CVaR penalty. The original LightGBM ensemble was not fitted because the information audit failed first.

The mean forecast is null. Ridge B0, 14 folds, 11,328 out-of-sample rows, 2010–2024: pooled OOS $R^2 = -0.019$, mean monthly rank IC 0.013, $t = 0.56$. A shuffled-target run gives mean IC −0.004, $t = -0.48$: the null is not an obvious leak.

The features do not survive. A date-blocked mutual-information audit over 259 cross-sections finds no feature that beats a permutation null in 80% of dates. The best pass rate is 0.552. LightGBM is not justified.

The reduced DRO does not separate. Table 1 reports the main result. The uncertainty-aware portfolio has a slightly higher Sharpe and a marginally smaller drawdown than point-estimate DRO, but the paired bootstrap intervals include zero for the tail differences. The 0.4% relative max-drawdown improvement is far below the pre-registered 20% threshold.

Strategy	Net Sharpe	Max DD	CVaR95	Turnover
M1 reduced	0.816	−24.18%	−8.31%	1.90
B5 point DRO	0.798	−24.28%	−8.44%	1.82
1/N	0.673	−26.61%	−9.76%	0.11

Table 1: v1 reduced DRO, 2010–2024, 179 months, net of 10 bps. M1 uses a Ridge mean and bootstrap sigma; it is not the original LightGBM ensemble.

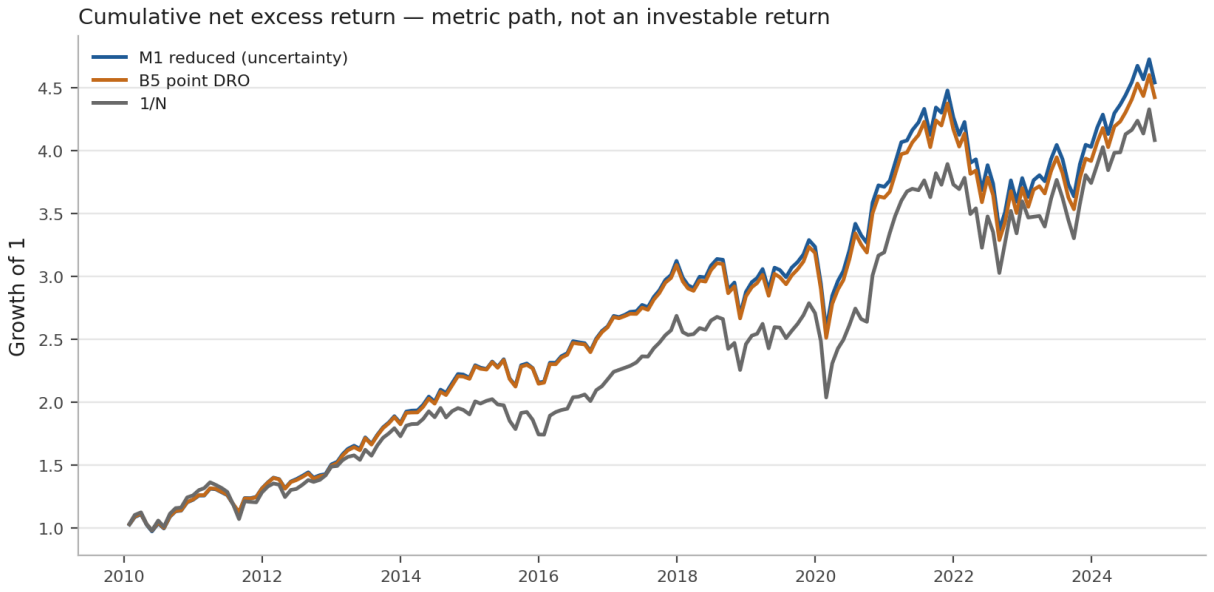


Figure 1: Cumulative net excess return for the three headline strategies. This is the metric's input path, not an investable return. The M1 and B5 paths are close; 1/N lags. The point is the absence of separation between M1 and B5, not the level.

Verdict. H1 fails: the tail improvement is 0.4% relative (90% CI [−6.2%, 1.1%]), and the CVaR difference interval includes zero. H2 passes: the Sharpe difference is +0.018, 90% CI [0.000, 0.037], inside the 0.20 margin. H3 fails: the uncertainty ablation is within noise. The one positive secondary result is that the DRO/CVaR optimizer beats 1/N on Sharpe and tail metrics; that is the optimizer, not the uncertainty layer.

4 v2: conformalized quantiles

The second attempt changed the object of uncertainty from bootstrap dispersion to a distributional forecast. LightGBM quantile regression predicts q05/q50/q95 of next-month excess return, three seeds are averaged, and split-conformal residual quantiles calibrate the interval on a 12-month validation window. The mean is the calibrated q50; sigma is the interval half-width divided by 1.645. The comparator is non-degenerate: each asset's trailing 36-month realized volatility. A constant sigma was rejected before the holdout because a constant penalty is weight-invariant and collapses to CVaR-only.

Calibration worked. Validation coverage was 89.9% development test coverage was 85.9%. But coverage is not information. The pre-holdout gate asked whether the conformal sigma beats a date-blocked permutation null in at least 80% of dates. It passed 65.4%. The holdout was not touched.

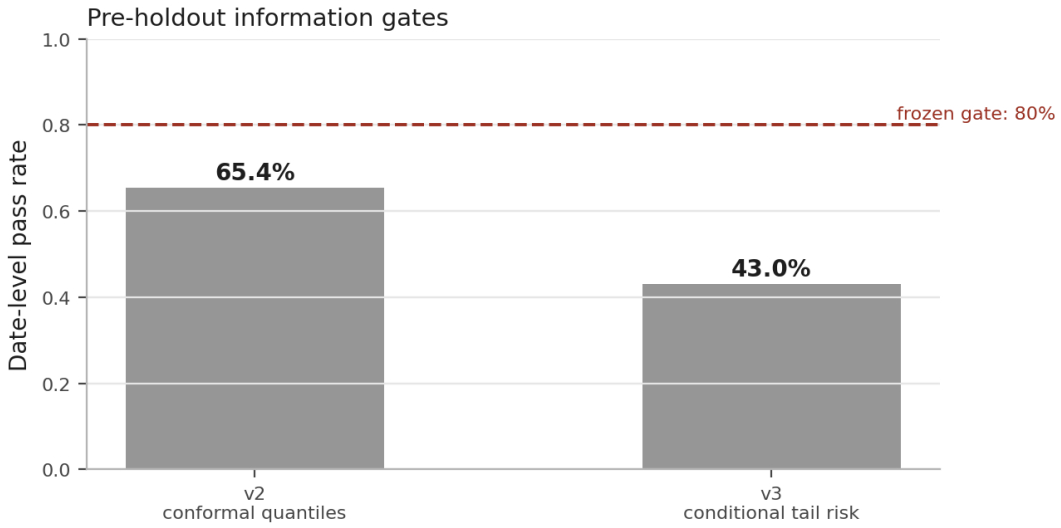


Figure 2: Pre-holdout information gates. v2 and v3 both fail the frozen 80% rule. The gate is deliberately strict: it prevents a calibrated but unstable signal from consuming the last untouched holdout.

5 v3: conditional tail risk

The third attempt changed the object again. Instead of epistemic uncertainty or a return quantile, it forecast **next-month absolute excess return** with LightGBM L1 regression and used the forecast as DRO sigma. This is a conditional tail-risk forecast, not epistemic uncertainty.

The development result is negative: M1 Sharpe 0.783, max drawdown −25.87%, CVaR95 −8.44%, against trailing volatility at 0.840, −24.76%, and −8.18%. The conditional forecast also trades more. It loses on every headline metric.

The calibration is the interesting part. Mean sigma is 4.29% against mean realized absolute return 4.30% the correlation is 0.28 the top-decile realized absolute return exceeds the bottom decile by 3.76%. A model can be well calibrated and cross-sectionally informative on average and still fail a stability gate. It passed 43.0% of date-level tests. The holdout was not touched.

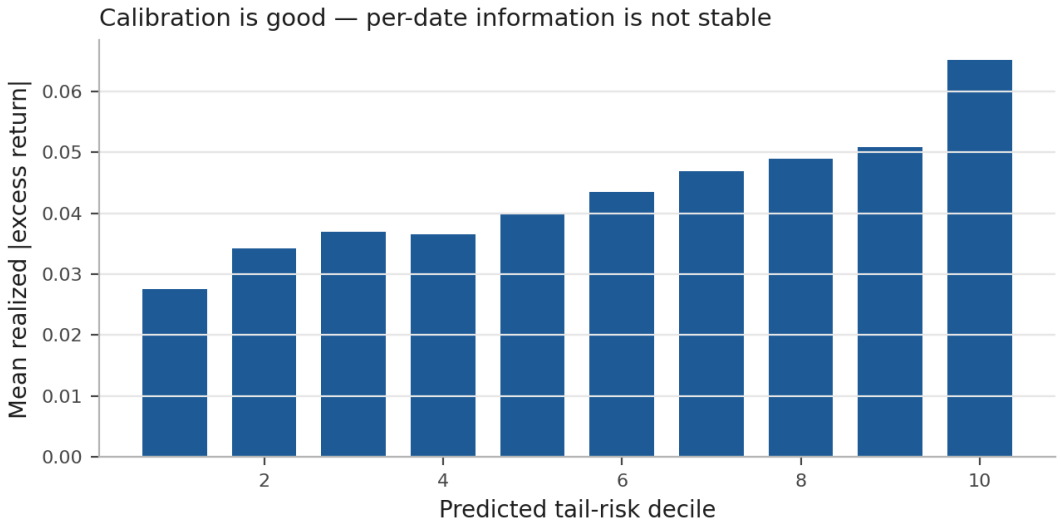


Figure 3: v3 predicted tail-risk deciles against mean realized absolute excess return. The monotone spread is real; the per-date information is not stable enough for the frozen gate.

6 Synthesis: why the answer is no

Three mechanisms explain the result.

The mean signal is weak. Ridge B0 has no detectable linear OOS signal, and the information audit finds no surviving feature. If the conditional mean is not predictable, the uncertainty layer has little to shrink. The binding constraint is the feature set, not the optimizer.

Uncertainty is easy to calibrate and hard to make informative. v2 and v3 are both calibrated. v3 has a clear decile spread. Neither passes a stability gate. Average calibration and cross-sectional spread are necessary but not sufficient; the signal must be stable date by date.

Comparison design matters. A constant sigma is a constant penalty in a long-only, sum-to-one box DRO; it does not change the optimizer. The first v2 comparator collapsed to CVaR-only for exactly this reason. The replacement — trailing realized volatility — is non-degenerate and is the right kind of benchmark: asset-specific, but not conditional on the model being tested.

The gates did their job. A project that only reports the final holdout would have consumed the 2025–2026 sample on a signal that failed its own pre-holdout test. Here the holdout is still available for a genuinely different model.

7 What would change the answer

Not a new feature search on the same data. A credible positive result needs a new pre-registration with:

1. a feature set that survives a pre-registered information audit;
2. an uncertainty model with a **stable** pre-holdout signal, not just good average calibration;
3. a non-degenerate comparator;
4. a new or untouched holdout; and
5. a decision rule fixed before the holdout is seen.

The most plausible next object is not epistemic uncertainty about the mean. It is a different target: tail risk, volatility, or a longer horizon where the signal-to-noise ratio is less brutal. v3 is the first step in that direction, and it failed for a specific, diagnosable reason.

8 Limitations

- The universe is current-listed; closed funds are omitted. The results are not survivorship-free.
- The ETF cross-section is narrower than a stock universe; power is limited.
- v1′s reduced M1 uses bootstrap Ridge sigma, not the original LightGBM ensemble.
- v2 and v3 never reached the holdout. Their negative results are development-period results.
- The v2/v3 holdout is only 20 months and cannot reliably detect a 20% relative MDD improvement.
- The conditional tail forecast is aleatoric, not epistemic; it is labelled as such.
- The linear-cost model is not a market-impact model. There is no live deployment.

9 Reproducibility

The code, frozen protocols, gates, and aggregate results are in the repository. Raw snapshots are gitignored. The main commands are:

```
python scripts/build_snapshot.py --force
python scripts/check_data.py
python scripts/run_baseline.py
python scripts/run_information_audit.py
python scripts/run_dro_experiment.py
python scripts/analyze_dro_experiment.py
python scripts/run_v2_experiment.py --period dev
python scripts/run_v2_uncertainty_audit.py
python scripts/run_v3_experiment.py --period dev
python scripts/run_v3_sigma_audit.py
python scripts/build_paper.py
pytest -q
ruff check .
```

The v2/v3 holdout commands are deliberately omitted: the frozen gates did not pass, so the holdout must not be run retrospectively.

10 References

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